Payment Service Setup Guide — CentOS 9

Payment service is responsible for payments in the RoboShop e-commerce application.

The original application documentation specifies Python 3.6. On CentOS 9, verify that the required Python 3.6 packages are available in your configured repositories before running the installation command.

1. Install Python and Required Packages

Install Python, GCC and Python development packages required by the Payment application.

```
dnf install python36 gcc python3-devel -y
```

2. Add Application User

Create the roboshop application user.

```
useradd roboshop
```

3. Create Application Directory

Create the application directory.

```
mkdir /app
```

4. Download Application Code

Download the Payment application artifact to /tmp.

```
curl -L -o /tmp/payment.zip https://roboshop-builds.s3.amazonaws.com/payment.zip
```

5. Extract Application Code

Change to /app and extract the downloaded application.

```
cd /app
unzip /tmp/payment.zip
```

6. Install Application Dependencies

Install the Python dependencies listed in requirements.txt.

```
cd /app
pip3.6 install -r requirements.txt
```

7. Configure SystemD Payment Service

Create /etc/systemd/system/payment.service and replace the placeholder server addresses with the actual Cart, User and RabbitMQ server IP addresses.

```
[Unit]
Description=Payment Service

[Service]
User=root
WorkingDirectory=/app
Environment=CART_HOST=<CART-SERVER-IPADDRESS>
Environment=CART_PORT=8080
```

```
Environment=USER_HOST=<USER-SERVER-IPADDRESS>
Environment=USER_PORT=8080
Environment=AMQP_HOST=<RABBITMQ-SERVER-IPADDRESS>
Environment=AMQP_USER=roboshop
Environment=AMQP_PASS=roboshop123

ExecStart=/usr/local/bin/uwsgi --ini payment.ini
ExecStop=/bin/kill -9 $MAINPID
SyslogIdentifier=payment

[Install]
WantedBy=multi-user.target
```

8. Reload SystemD

Reload systemd so it recognizes the newly created Payment service.

```
systemctl daemon-reload
```

9. Enable and Start Payment Service

Enable Payment to start automatically at boot and start the service.

```
systemctl enable payment
systemctl start payment
```

10. Verify Payment Service

Check the current status of the Payment service.

```
systemctl status payment
```

Final Command Sequence

```
dnf install python36 gcc python3-devel -y
useradd roboshop
mkdir /app
curl -L -o /tmp/payment.zip https://roboshop-builds.s3.amazonaws.com/payment.zip
cd /app
unzip /tmp/payment.zip
pip3.6 install -r requirements.txt

# Create /etc/systemd/system/payment.service
systemctl daemon-reload
systemctl enable payment
systemctl start payment

systemctl status payment
```

Important: Replace <CART-SERVER-IPADDRESS>, <USER-SERVER-IPADDRESS> and <RABBITMQ-SERVER-IPADDRESS> with the actual server addresses. The RabbitMQ credentials must match the user created on the RabbitMQ server.